

Question:

Design a “farm animals” Java application with the details of animals like cow, pig, horse. Consider the following details: where they stay, what they eat, and the sound they make, using classes and objects.

Steps for Execution

1. Open any Java IDE (e.g., Eclipse, IntelliJ, or BlueJ) or a text editor.
2. Create a new Java file named FarmAnimals.java.
3. Copy the given code into the file.
4. Compile the program (javac FarmAnimals.java).
5. Run the program (java FarmAnimals).
6. Observe the output printed in sentence form.

Algorithm

1. Define a class Animal with attributes: name, place, food, and sound.
2. Create a constructor to initialize these attributes.
3. Define a method displayDetails() to print the animal’s information in sentence form.
4. In the FarmAnimals main class:
 - Create objects for Cow, Pig, Horse, Goat, Duck, and Chicken.
 - Call displayDetails() for each object.
5. End program.

UML Class Diagram

```
+-----+
|  Animal  |
+-----+
| - name: String |
| - place: String |
```

```

| - food: String |
| - sound: String |
+-----+
| + Animal(...) |
| + displayDetails()|
+-----+

FarmAnimals (main class)

|
| creates objects
v
Cow, Pig, Horse,
Goat, Duck, Chicken

```

Code:

```

// Base class
class Animal {
    String name;
    String place;
    String food;
    String sound;

    Animal(String name, String place, String food, String sound) {
        this.name = name;
        this.place = place;
        this.food = food;
        this.sound = sound;
    }
}

```

```

void displayDetails() {
    System.out.println("The " + name + " lives in the " + place +
        ", eats " + food +
        ", and makes the sound \"" + sound + "\".");
}
}

public class FarmAnimals {
    public static void main(String[] args) {
        Animal cow = new Animal("Cow", "cowshed", "grass & hay", "Moo Moo");
        Animal pig = new Animal("Pig", "pigsty", "vegetables & grains", "Oink Oink");
        Animal horse = new Animal("Horse", "stable", "hay & oats", "Neigh Neigh");
        Animal goat = new Animal("Goat", "goat pen", "leaves & grass", "Baa Baa");
        Animal duck = new Animal("Duck", "pond", "insects & grains", "Quack Quack");
        Animal chicken = new Animal("Chicken", "henhouse", "grains & seeds", "Cluck
        Cluck");

        cow.displayDetails();
        pig.displayDetails();
        horse.displayDetails();
        goat.displayDetails();
        duck.displayDetails();
        chicken.displayDetails();
    }
}

```

Output Screenshot:

The Cow lives in the cowshed, eats grass & hay, and makes the sound "Moo Moo".

The Pig lives in the pigsty, eats vegetables & grains, and makes the sound "Oink Oink".

The Horse lives in the stable, eats hay & oats, and makes the sound "Neigh Neigh".

The Goat lives in the goat pen, eats leaves & grass, and makes the sound "Baa Baa".

The Duck lives in the pond, eats insects & grains, and makes the sound "Quack Quack".

The Chicken lives in the henhouse, eats grains & seeds, and makes the sound "Cluck Cluck".